Distributed Word Representations: A Comparative Analysis of Skip-gram and GloVe

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Abstract

This paper provides a comparative analysis of two classical word embedding models in natural language processing. Skip-gram and GloVe. The paper explores how each model represents semantic meaning, examines their training objectives and assumptions, and evaluates their performance on standard analogy tasks. While Skip-gram emphasizes local context prediction and incremental training, GloVe leverages global co-occurrence statistics for stable, interpretable embeddings. The discussion reflects on the theoretical and practical implications of each model and considers how their insights continue to influence modern language models.

1 Introduction

1.1 Motivation: Learning Word Representations

Understanding the meaning of words based on their usage in text is a fundamental challenge in natural language processing (NLP). Traditional approaches, such as **count-based methods** or **one-hot encoding**, failed to capture deeper semantic relationships between words. To address these limitations, neural network-based **word embeddings** emerged as an effective method to represent words in a continuous vector space, where words with similar meanings are closer together.

One of the most influential advances in this area is the **skip-gram model**, introduced by Mikolov et al. (2013). This model efficiently learns high-dimensional word vectors by training a neural network to predict context words given a target word. The resulting embeddings encode meaningful relationships between words, making them useful for various NLP applications.

1.2 Importance

Word embeddings, such as those produced by the Skip-gram and GloVe models, represent an important step in the evolution of natural language understanding. Although more recent advances, such as contextual embeddings from transformer-based models like Gemini and GPT, have further improved performance in downstream tasks, these newer systems still build upon the core idea that meaning can be encoded in high-dimensional vector spaces. This paper focuses specifically on classical word embeddings, which remain foundational and are essential for understanding the development of modern NLP.

While transformer-based models Vaswani et al. (2017) now dominate NLP, their design inherits core ideas from static embeddings:

- The notion of *distributed representations* (from Skip-gram/GloVe) persists in transformers' input embeddings.
- *Vector space geometry* for semantic relationships (e.g., analogy tasks) directly inspired transformer self-attention mechanisms.

1.3 Contributions of the Paper

The key contributions of Mikolov et al.'s work can be summarized as follows:

- **The Skip-gram Model:** A neural-based method to learn word representations that outperforms previous approaches in terms of accuracy and efficiency.
- **Vector Arithmetic in Word Embeddings:** The discovery that simple vector operations (e.g. *king* - *man* + *woman* = *queen*) capture meaningful linguistic relationships.
- **Negative Sampling:** A computationally efficient alternative to hierarchical softmax for training word embeddings.
- **Subsampling of Frequent Words:** A technique that improves training efficiency and improves the representation of rare words.
- **Phrase Representations:** A method for extending word embeddings to multi-word phrases, improving the expressiveness of the model.

1.4 Comparison to Other Methods

Although Skip-gram is highly effective, other models have also been proposed for learning word representations. One notable alternative is **GloVe (Global Vectors for Word Representation)** (Pennington et al. 2014), which constructs word vectors by analyzing co-occurrence statistics rather than predicting context words. While Skip-gram is optimized for learning from local contexts, GloVe captures global statistical relationships.

1.5 Structure of the Essay

The rest of this paper is organized as follows:

- **Section 2** provides a detailed explanation of the Skip-gram model, including its mathematical formulation and optimizations.
- **Section 2.6** summarizes the experimental results reported by Mikolov et al.
- **Section 3** introduces the GloVe model and explains its theoretical foundations and performance.

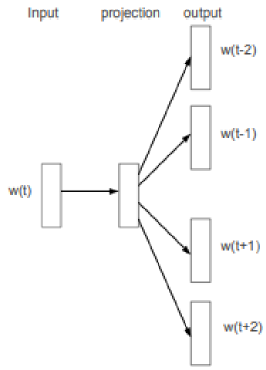

Figure 1: The Skip-gram model architecture. The model learns to predict words in a context window given a center word. Source: Mikolov et al. (2013).

- **Section 4** presents a comparative analysis of Skip-gram and GloVe.
- **Section 5** offers a critical reflection on the assumptions, limitations, and broader implications of the models.
- **Section 6** summarizes the key findings and reflects on future directions.

2 The Skip-gram Model

The Skip-gram model, introduced by Mikolov et al. (2013), is a predictive model that learns word embeddings by optimizing the ability to predict context words from a given target word. The central idea is that the meaning of a word can be inferred from the words that surround it. Unlike traditional count-based methods, Skip-gram constructs word vectors based on local co-occurrence patterns rather than entire dataset-wide statistics.

As illustrated in Figure 1, the model takes a single input word and is trained to predict each word in a surrounding window. The learned embeddings are the parameters that optimize this prediction task.

2.1 Training Objective

The model is trained to maximize the probability of a context window around each word in the dataset. Formally, given a sequence of training words $w_1, w_2, \ldots, w_T$, the Skip-gram objective function is to maximize the average log-probability:

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{-c\leq j\leq c, j\neq 0}\log p(w_{t+j}|w_t),$$

where c is the context window size. The conditional probability $p(w_O|w_I)$ of a context word w_O given an input word w_I is modeled using the softmax function:

$$p(w_O \mid w_I) = \frac{\exp\left(\mathbf{v}'{w_O}^\top \mathbf{v}{w_I}\right)}{\sum_{w=1}^{W} \exp\left(\mathbf{v}'{w}^\top \mathbf{v}{w_I}\right)}$$

where $\mathbf{v}_w$ and $\mathbf{v}'_w$ are the input and output vector representations of word w, and W is the vocabulary size. Since the scaling factor involves a sum over the entire vocabulary, computing the gradient is expensive for large datasets.

2.2 Hierarchical Softmax

To reduce computational cost, Mikolov et al. (2013) propose the use of **hierarchical softmax**, a binary tree-based approximation of the full softmax. Each word is a leaf node, and the probability of a word is computed as the probability of a path from the root to that word. For a word w, let $n(w, j)$ be the j-th node along the path from root to w, and $L(w)$ be the path length. Then:

$$p(w|w_I) = \prod_{j=1}^{L(w)-1} \sigma\left([n(w, j+1) = \mathrm{ch}(n(w, j))] \cdot {\mathbf{v}'_{n(w,j)}}^\top \mathbf{v}_{w_I} \right),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function and the Iverson bracket returns 1 if the condition is true, and -1 otherwise. This reduces the complexity from $O(W)$ to $O(\log W)$.

2.3 Negative Sampling

An alternative to hierarchical softmax is **Negative Sampling**, which reframes the problem as a binary classification: distinguishing real context word pairs from randomly sampled "negative" ones. The simplified training objective becomes:

$$\log \sigma(\mathbf{v}'_{w_O} \cdot \mathbf{v}_{w_I}) + \sum_{i=1}^{k} \mathbf{E}_{w_i \sim P_n(w)} \left[\log \sigma(-{\mathbf{v}'_{w_i}}^\top \mathbf{v}_{w_I}) \right],$$

where k is the number of negative samples, and $P_n(w)$ is a noise distribution. This method is highly efficient and particularly effective for frequent words.

2.4 Subsampling of Frequent Words

Frequent words such as "the", "and", or "in" provide little semantic information but appear excessively in large datasets. The model implements **subsampling** to discard such words with a probability:

$$P(w_i) = 1 - \sqrt{\frac{t}{f(w_i)}},$$

where $f(w_i)$ is the frequency of word w_i, and t is a threshold (typically 10^{-5}). This reduces the imbalance between frequent and rare words, accelerates training, and improves representation quality.

Table 1: Skip-gram accuracy on the word analogy task (source: Mikolov et al. (2013))

Method	Syntactic (%)	Semantic (%)	Total Accuracy (%)
NEG-5	63	54	59
NEG-15	63	58	61
HS-Huffman	53	40	47
NCE-5	60	45	53
NEG-5 (subsampled)	61	58	60
NEG-15 (subsampled)	61	61	61
HS-Huffman (subsampled)	52	59	55

2.5 Phrase Representations and Compositionality

The Skip-gram model can be extended to phrases by identifying frequent multi-word expressions and treating them as single tokens. This makes the model more expressive and enables it to capture meanings that are not compositional in nature (e.g., "New York Times", "Air Canada"). Furthermore, Skip-gram embeddings exhibit a surprising property: linear vector operations often encode meaningful linguistic analogies such as:

$$\text{vec}(''\textit{king}'') - \text{vec}(''\textit{man}'') + \text{vec}(''\textit{woman}'') \approx \text{vec}(''\textit{queen}'').$$

This linear structure forms the basis of many downstream applications.

2.6 Experimental Results

Mikolov et al. evaluate the Skip-gram model on an analogical reasoning task, where questions are posed in the form: "a is to b as c is to ?". The task is split into two categories: **semantic analogies** (e.g., "France is to Paris as Germany is to ?") and **syntactic analogies** (e.g., "walking is to walked as swimming is to ?").

Table 1 summarizes the performance of different Skip-gram variants using 300-dimensional embeddings trained on a dataset of 1 billion words. Accuracy is measured based on how often the closest vector to the expected analogy result is the correct answer.

The results show that:

- **Negative sampling** consistently outperforms both hierarchical softmax and noise contrastive estimation.
- **Subsampling of frequent words** improves both training efficiency and the quality of embeddings, particularly for rare words.
- The highest accuracy (61%) is obtained with NEG-15 and subsampling, demonstrating the effectiveness of these two strategies combined.

These empirical findings validate the hypothesis that training efficiency and embedding quality can be simultaneously improved with suitable optimizations.

To further illustrate the semantic structure captured by the model, Figure 2 shows a two-dimensional PCA projection of the Skip-gram embeddings. The

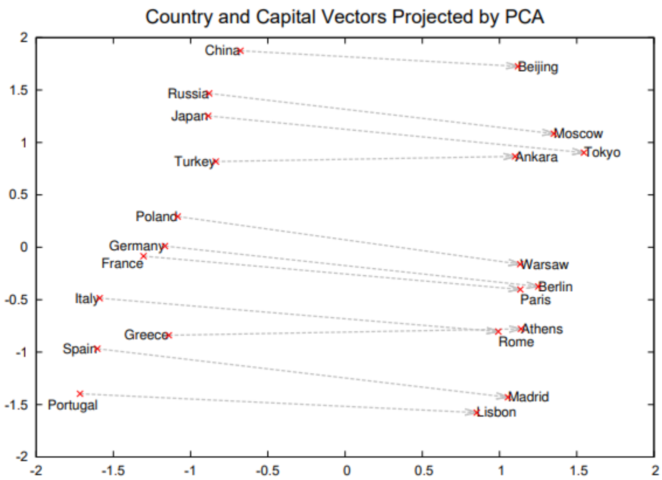

Figure 2: Two-dimensional PCA projection of 1000-dimensional Skip-gram embeddings of countries and their capitals. Despite no supervision during training, the model organizes semantically related entities into structured regions. Source: Mikolov et al. (2013).

figure demonstrates that countries and their corresponding capital cities are organized into meaningful geometric relationships in the vector space.

3 GloVe: A Global Approach

While the Skip-gram model focuses on predicting surrounding words in a local window, **GloVe (Global Vectors for Word Representation)** offers a different approach. Proposed by Pennington et al. (2014), the model integrates the benefits of count-based and predictive models by learning word vectors directly from co-occurrence statistics aggregated across the entire dataset.

3.1 Motivation and Intuition

The authors of GloVe argue that meaningful semantic relationships arise not from individual co-occurrences alone, but from the ratios of co-occurrence probabilities between words. For example, the relationship between "ice" and "steam" can be better understood by comparing how often they co-occur with a set of context words like "solid" and "gas":

$$\frac{P(\text{solid}|\text{ice})}{P(\text{solid}|\text{steam})} \gg 1, \quad \frac{P(\text{gas}|\text{ice})}{P(\text{gas}|\text{steam})} \ll 1,$$

These ratios reveal fine-grained distinctions in meaning that are otherwise hard to capture using only local context windows.

3.2 The GloVe Model

Let $X \in \mathbf{R}^{V \times V}$ be a matrix of word-word co-occurrence counts, where X_{ij} indicates how often word j appears in the context of word i. The goal is to learn word vectors $\mathbf{w}_i$ and context vectors $\tilde{\mathbf{w}}_j$ such that their dot product approximates the logarithm of their co-occurrence count:

$$\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j = \log(X_{ij}),$$

where b_i and $\tilde{b}_j$ are bias terms for word and context respectively.

To avoid giving too much weight to extremely frequent or infrequent co-occurrences, the model introduces a weighting function $f(X_{ij})$, resulting in the following cost function:

$$J = \sum_{i,j=1}^{V} f(X_{ij}) \left(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log(X_{ij}) \right)^2 .$$

The weighting function is defined as:

$$f(x) = \begin{cases} (x/x_{\max})^\alpha & \text{if } x < x_{\max}, \\ 1 & \text{otherwise,} \end{cases}$$

where $x_{\max}$ and α are tunable hyperparameters. The authors recommend $x_{\max} = 100$ and $\alpha = 3/4$, which have been empirically validated.

3.3 Theoretical Comparison with Skip-gram

Whereas Skip-gram builds word vectors by predicting surrounding words one pair at a time, GloVe optimizes its vectors to reflect the **global structure** of the dataset in a single objective function. GloVe can be interpreted as performing an implicit factorization of the log-co-occurrence matrix. This gives it two key advantages:

- It preserves global statistics and regularities that are missed by local window-based approaches.
- It directly encodes semantic relationships via co-occurrence *ratios*, which can be interpreted as directions in the embedding space.

Despite these differences, both models yield vector spaces in which semantic relationships can be recovered with an linear vector arithmetic. In fact, both demonstrate impressive performance on analogy tasks, suggesting that a shared geometric structure underlies both learning paradigms.

3.4 Empirical Performance

GloVe achieves competitive or superior results compared to Skip-gram across various benchmarks. On the analogy task, a 300-dimensional GloVe model trained on 6 billion tokens achieved 71.7% accuracy, outperforming Skip-gram's 69.1% under comparable settings (Pennington et al. 2014). Moreover, GloVe vectors have shown state-of-the-art performance on word similarity and named entity recognition tasks.

Although GloVe is less flexible during incremental updates (since it requires precomputing a co-occurrence matrix), its training process is highly parallelizable and efficient for large static datasets. This makes it particularly well-suited for industrial-scale applications. GloVe performs especially well on large, static datasets due to its use of full co-occurrence statistics. However, its need for a precomputed matrix limits its flexibility in dynamic or streaming data contexts, where Skip-gram has an advantage.

4 Comparative Analysis

The Skip-gram and GloVe models represent two complementary frameworks in learning word embeddings. While both aim to capture semantic and syntactic relationships through vector representations, they differ significantly in terms of training strategy, underlying assumptions, and practical performance. This section presents a comparative analysis across several dimensions.

4.1 Local Prediction vs. Global Co-occurrence

At a high level, Skip-gram is a predictive model that uses a sliding context window to learn embeddings by predicting context words given a center word. In contrast, GloVe is a count-based model that relies on the full co-occurrence matrix, optimized via global least-squares regression.

- **Skip-gram** learns embeddings from observed context pairs: each word pair (target, context) contributes one training sample.
- **GloVe** considers all word pairs with non-zero co-occurrence, including pairs not directly adjacent in the text.

Thus, Skip-gram captures local structure well, while GloVe incorporates global dataset statistics into its representation.

4.2 Efficiency and Scalability

Skip-gram is extremely efficient and well-suited for streaming or online training. The use of negative sampling and subsampling of frequent words allows the model to be trained on very large datasets in linear time.

GloVe, in contrast, requires computing and storing a co-occurrence matrix beforehand. While this allows better parallelization and matrix factorization-style efficiency, it also introduces memory constraints for extremely large vocabularies.

- **Skip-gram**: optimized for fast, incremental learning on sequential data.
- **GloVe**: optimized for static, large-scale dataset where full co-occurrence statistics are available.

4.3 Interpretability and Theoretical Motivation

GloVe is motivated by the idea that word meaning can be captured through ratios of co-occurrence probabilities. The resulting model is grounded in matrix factorization theory, and its cost function reflects a clear mathematical structure. Its vectors are trained to reconstruct the logarithm of observed co-occurrence counts, producing dimensions that align with interpretable directions in semantic space.

Skip-gram, on the other hand, is less theoretically motivated but highly empirical. Its learning objective, predicting context words from a center word, relies more on engineering decisions such as negative sampling and sub-sampling. Rather than optimizing a formal statistical objective over a global matrix, it operates locally and gradually.

This contrast reflects a broader tension in the field of distributional semantics. While models like GloVe offer theoretical elegance, predictive models like Skip-gram have demonstrated strong empirical results. Baroni et al. (2014) argue that prediction-based approaches, although lacking a well established theoretical foundation, often outperform count-based models on real-world NLP tasks. This raises the question of whether formal justification is always necessary when empirical performance is the primary goal.

4.4 Performance on Benchmarks

On the word analogy benchmark introduced by Mikolov et al. (2013), both models achieve competitive results. As shown by Pennington et al. (2014), GloVe with 300-dimensional vectors trained on 6B tokens achieves 71.7% accuracy, slightly outperforming Skip-gram's 69.1%.

Although GloVe's 2.6% higher accuracy on the analogy task suggests better global consistency, Mikolov et al. note that Skip-gram's performance gap narrows in tasks requiring incremental updates (e.g., streaming data). This implies that GloVe's advantage may be dependent on the dataset.

Moreover, GloVe embeddings tend to perform better on word similarity tasks and named entity recognition (NER), while Skip-gram tends to excel in low-resource settings and tasks that benefit from gradual updates.

4.5 Compositionality and Vector Arithmetic

Both models exhibit a striking property: linear relationships between vectors often correspond to meaningful linguistic transformations. This has been demonstrated through analogy tasks such as:

$$\mathrm{vec}(\text{"}king\text{"}) - \mathrm{vec}(\text{"}man\text{"}) + \mathrm{vec}(\text{"}woman\text{"}) \approx \mathrm{vec}(\text{"}queen\text{"}),$$

Although Skip-gram was the first to reveal this phenomenon clearly, GloVe embeddings exhibit similar compositional behavior. This suggests that both models induce a latent semantic structure in the embedding space, despite their different training regimes.

Table 2: Comparison of Skip-gram and GloVe

Aspect	**Skip-gram**	**GloVe**
Training objective	Predict context words using local window	Approximate log co-occurrence counts using matrix factorization
Learning paradigm	Predictive (online)	Count-based (batch)
Scope	Local (sliding window)	Global (full dataset co-occurrence)
Optimization method	Stochastic (Negative Sampling / HS)	Weighted least-squares regression
Scalability	Efficient for large, streaming datasets	Requires precomputed co-occurrence matrix
Interpretability	Empirical, task-driven	Theoretically motivated via ratios
Compositionality	Emerges from prediction	Emerges from co-occurrence ratios

4.6 Summary of Differences

In conclusion, while the Skip-gram and GloVe models differ significantly in methodology, both succeed in capturing complex linguistic patterns through distributed vector representations. Their complementary strengths suggest that the choice between them depends on the specific task and computational constraints.

5 Discussion and Reflection

Reflecting on the contributions of Mikolov et al. (Mikolov et al. 2013) and Pennington et al. (Pennington et al. 2014), I am struck by how differently these two papers approach the same fundamental task, learning meaningful word representations from raw text, and yet arrive at embeddings that perform similarly across many applications.

5.1 A Comparison of Intuition

The Skip-gram model is built around a very simple idea: if a word can help predict its surrounding words, it must encode something about their meaning. What is powerful here is not the model's architecture, which is minimal, but the fact that this prediction task, executed billions of times, yields a vector space where structure emerges naturally. In my view, this is both fascinating and a bit counterintuitive: a shallow prediction task trained with negative examples and noise can uncover syntactic and semantic patterns.

GloVe, by contrast, starts from the idea that meaning is inherently relational and statistical. The use of co-occurrence ratios is a more principled way to model meaning. It appeals to linguistic theories that stress distributional context, and it connects more explicitly to traditional matrix factorization. From a theoretical standpoint, GloVe is more elegant. But this elegance does not always translate into practical superiority.

Interestingly, Levy and Goldberg (2014) argue that Skip-gram with negative sampling is implicitly factorizing a shifted pointwise mutual information (PMI) matrix. This finding suggests that the boundary between predictive and count-based models may not be as clear-cut as it seems. Despite their conceptual differences, both models end up capturing similar statistical relationships, just through different means.

5.2 On Simplicity vs. Formalism

One of the most interesting contrasts between the two papers is the role of formalism. Mikolov et al. present an empirical approach with little mathematical justification. Their innovations, such as negative sampling and subsampling, are guided by engineering insight, not theory. And yet, their results are impressive. GloVe, in contrast, justifies every step, from the form of the cost function to the weighting scheme.

This leads me to reflect on the role of theory in machine learning. GloVe's log-bilinear formulation is satisfying on paper, but in practice, Skip-gram performs similarly or better in some tasks. It makes me question whether theoretical elegance is always necessary, especially when data and computation are abundant.

5.3 Critical Questions about Evaluation

Both papers rely heavily on the analogy task as a benchmark, where vector arithmetic like:

$$\text{vec}(''king'') - \text{vec}(''man'') + \text{vec}(''woman'') \approx \text{vec}(''queen'')$$

serves as a sign of successful learning. While this task is clever and intuitive, I find it somewhat misleading. Language is more complex than analogies: irony, context, pragmatics, and world knowledge are not captured by vector differences. The analogy benchmark reduces meaning to geometry, which may say more about our evaluation limitations than about the model's capabilities.

In this light, I believe that both papers slightly overstate the usefulness of these vector properties. While they are impressive, especially when visualized or applied in clustering, they don't fully reflect the richness of linguistic understanding.

5.4 Practical Reflections

From a practical perspective, I find Skip-gram easier to work with. Its streaming-friendly nature and straightforward architecture make it more adaptable to low-resource or dynamic environments. I've experimented with it in small NLP projects, and the ability to train on-the-fly and control the quality through hyperparameters like context window and negative samples is a major advantage.

GloVe, by contrast, requires an up-front investment in computing the co-occurrence matrix. For large datasets, this can be a bottleneck. While its training objective is clean and differentiable, the preprocessing makes it harder to integrate into dynamic pipelines.

5.5 Concluding Reflection

In comparing the two, I find myself more impressed by the pragmatism of Skip-gram than the theory of GloVe. Yet, I acknowledge the value of both. GloVe's approach feels more linguistically grounded, while Skip-gram feels more like a hack that just happens to work extremely well. It's tempting to view GloVe as the "ideal" and Skip-gram as the "practical," but in truth, both models reflect different paths to the same goal: making human language computable.

Looking back, these two papers represent not just technical contributions, but two philosophies of machine learning empiricism versus theory. The most important lesson, for me, is that good representations can come from either, and that it's the alignment of model, data, and task that ultimately matters most.

6 Conclusion

Key Findings

This essay has explored and analyzed two of the most influential models for learning distributed word representations: the Skip-gram model by Mikolov et al. and the GloVe model by Pennington et al. Despite their differences in methodology, one based on local prediction, the other on global co-occurrence, they both demonstrate that semantic and syntactic regularities in language can be effectively captured through geometric representations in vector space.

The Skip-gram model stands out for its simplicity, scalability, and ability to learn high-quality embeddings through efficient sampling techniques. GloVe, in contrast, offers a theoretically grounded framework that leverages the statistical structure of the entire dataset. Both models exhibit the fascinating property of vector compositionality, allowing for meaningful analogical reasoning and downstream applications.

The comparative analysis shows that no single approach dominates across all dimensions. Skip-gram may be preferable in streaming or dynamic settings, while GloVe is advantageous when global dataset statistics are available and memory is less constrained. Ultimately, the choice depends on the use case.

Modern impact and Legacy

Several open questions remain. Among them is how to move beyond static word representations toward embeddings that capture context, meaning shifts, and pragmatic implications. While modern transformer-based models Vaswani et al. (2017) address these limitations through dynamic attention mechanisms, their design inherits key insights from Skip-gram and GloVe:

- **From co-occurrence to attention**: GloVe's global co-occurrence statistics are generalized by transformers' self-attention heads, which weight context dynamically based on learned pairwise relationships.
- **From sliding windows to hierarchical context**: Skip-gram's local windowed prediction is extended to variable-length dependencies via transformer layers, while preserving the geometric intuition of vector-space semantics.

Although contextual models (e.g., Gemini, GPT) dominate modern NLP, the conceptual foundations of static embeddings persist: transformers still initialize with pretrained word embeddings, and their attention mechanisms refine, rather than replace, the distributional hypotheses formalized by Skip-gram and GloVe.

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